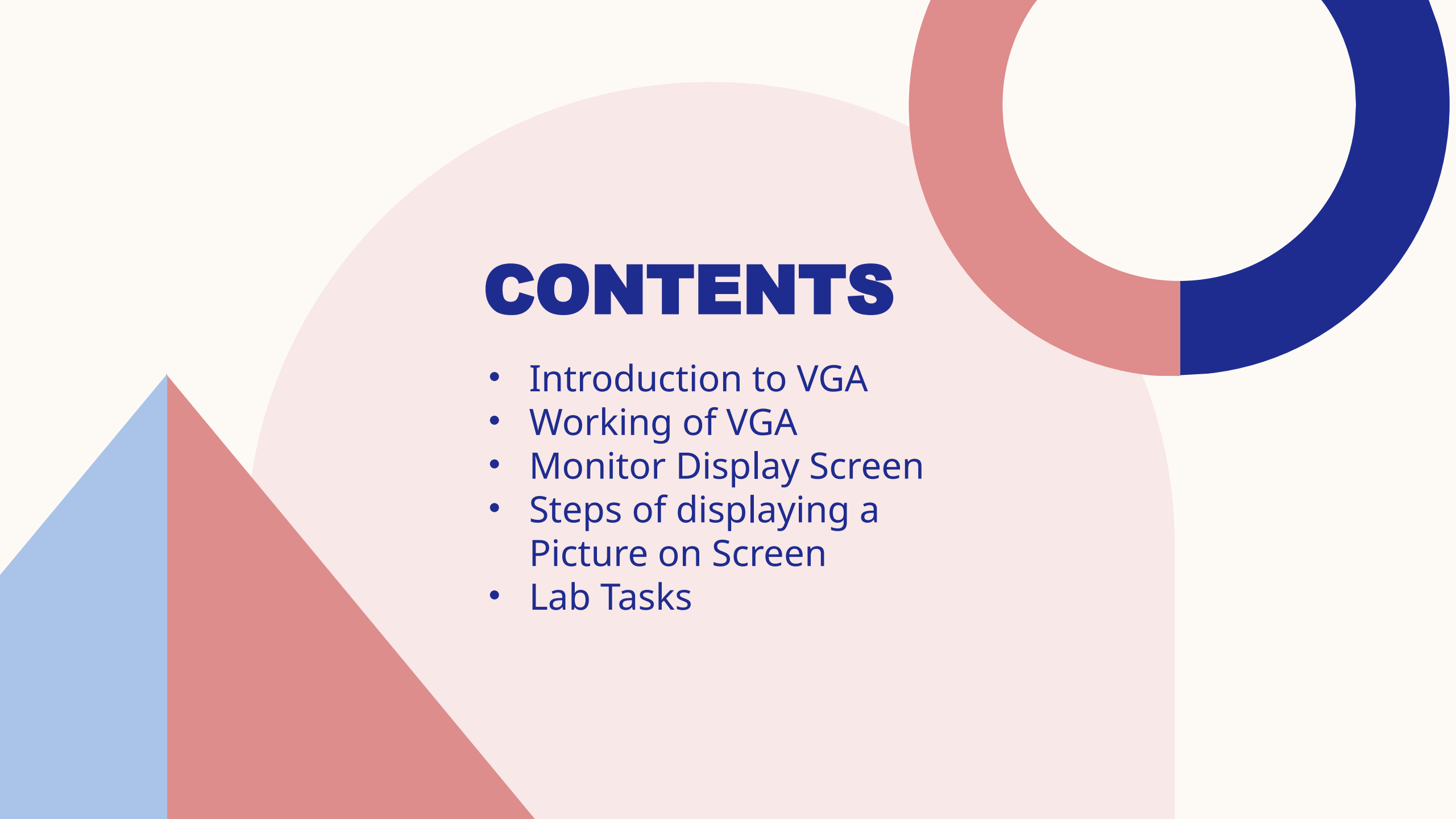


LAB 11

VGA CONTROLLER

**RA: Ms. Maham
Tabassum**



CONTENTS

- Introduction to VGA
- Working of VGA
- Monitor Display Screen
- Steps of displaying a Picture on Screen
- Lab Tasks

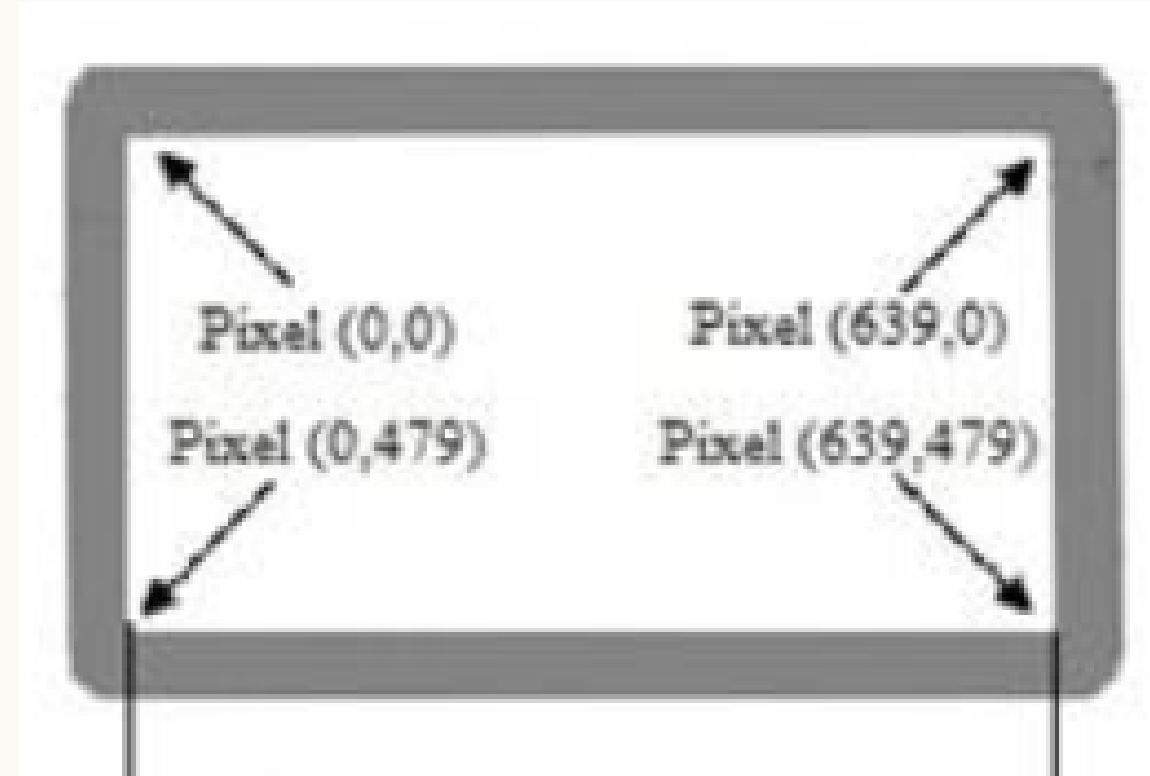
VGA - VIDEO GRAPHICS ARRAY

- VGA is a display standard used to interface digital systems, such as FPGAs, microcontrollers, or computers, with a monitor.
- It is an interface protocol used to transmit analog video data to a screen. It allows graphical or text-based output to be displayed on a screen.
- VGA connectors are used with televisions, laptops, computer monitors, projectors, and other devices.
- **Display resolution of 640 x 480 (h x v) pixels.**



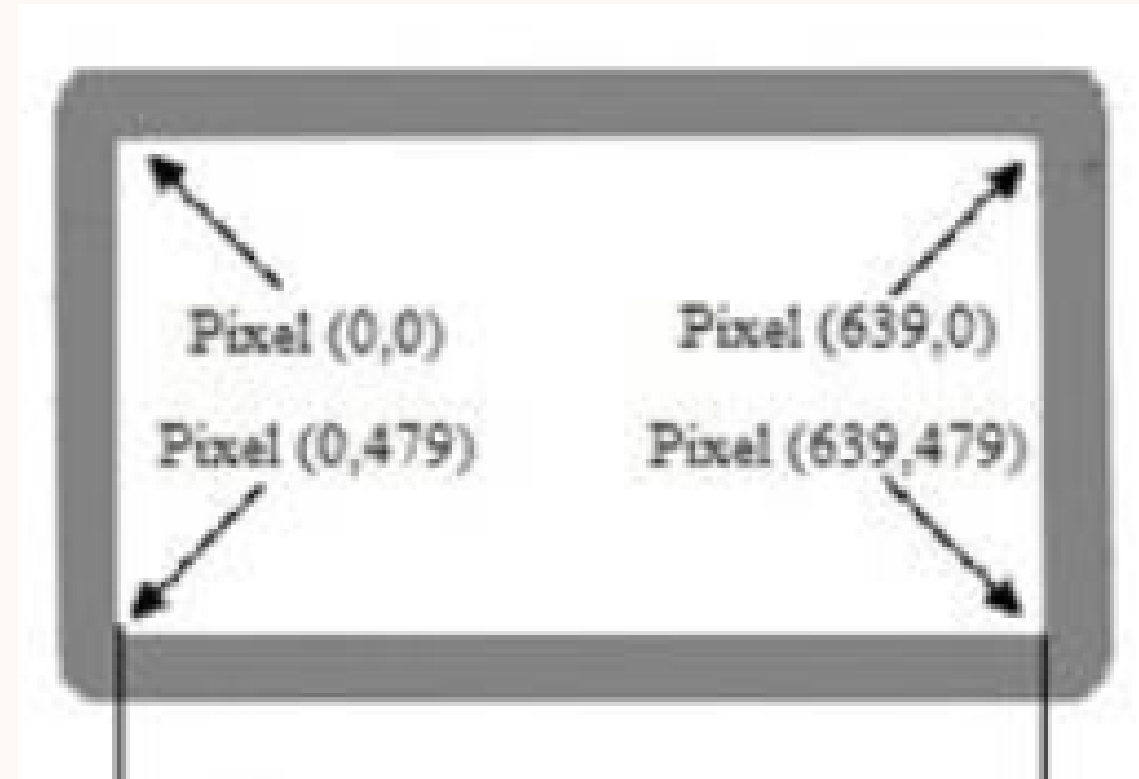
MONITOR SCREEN

- The screen of a monitor includes:
 1. Small black border is the outer covering of the monitor. The lengths of the borders may vary for different brands of monitors.
 2. Middle rectangle – Display Surface/Visible Portion is the region where the pixels are actually displayed on the screen.



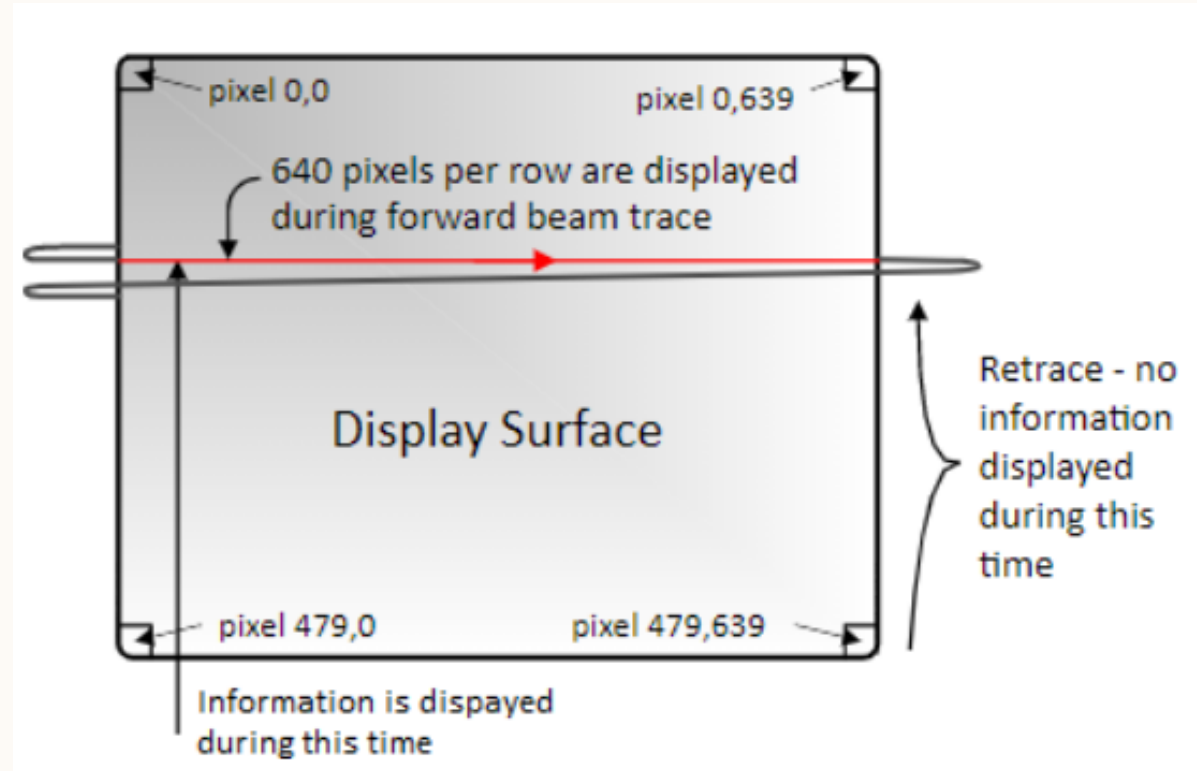
MONITOR DISPLAY SCREEN

- The pixel range of the visible portion of the screen:
 1. Horizontal (640 pixels): 0 – 639 (increasing towards right).
 2. Vertical (480 pixels): 0 - 479 (increasing downwards).



WORKING OF VGA CONTROLLER

- To project an image, VGA uses a scanning method using an electron beam that scans the screen row by row.
- Each row is scanned pixel by pixel.
- Starting at the top-left corner of the screen, beam progressively moves from left to right (*pixel by pixel*), top to bottom (*row by row*) to display each pixel.
Beam starts from top-left pixel and ends at the bottom-right pixel.



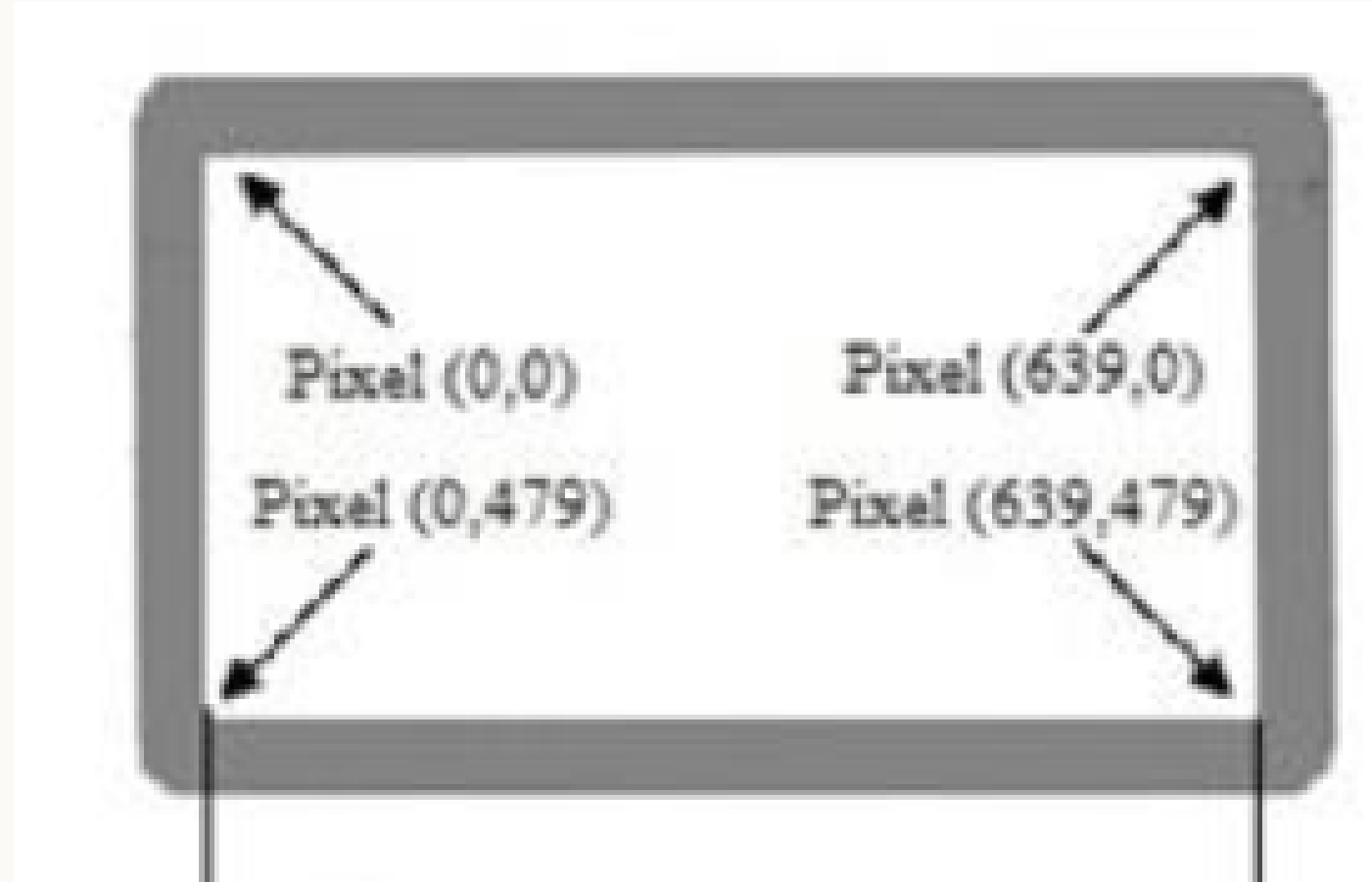
WORKING OF VGA CONTROLLER

- This beam moves using two synchronization signals, called *h_sync* (horizontal synchronization), and *v_sync* (vertical synchronization).
 1. The *h_sync* signal tells the beam when to move to the next row.
 2. The *v_sync* signal tells the screen that the current video frame is completed. The beam then moves back to the top of the screen for the new frame.
- Apart from these two signals, following three signals must also be sent to VGA Monitor in order to display the image:
Red, Green, Blue (RGB) - three separate analog voltage signals indicating the amount of each color to display in the current pixel.

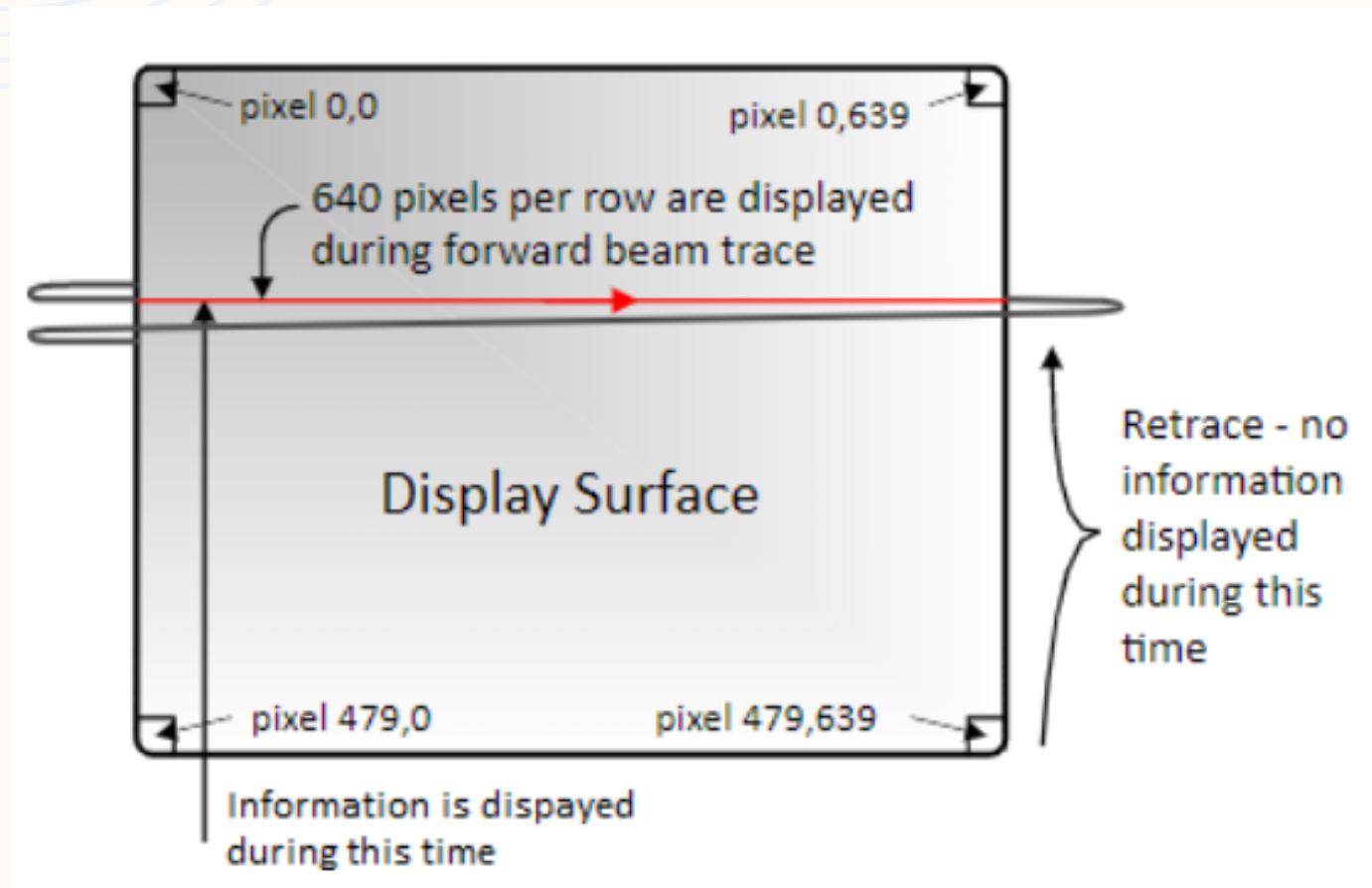
VGA BEAM SCAN

- Horizontal Scan
 1. Right border (Front Porch)
 2. Retrace
 3. Left border (Back Porch)
- Vertical Scan
 1. Bottom border (Front Porch)
 2. Retrace
 3. Top border (Back Porch)

Front porch: Porch before retrace
Back porch: Porch after retrace



RETRACE



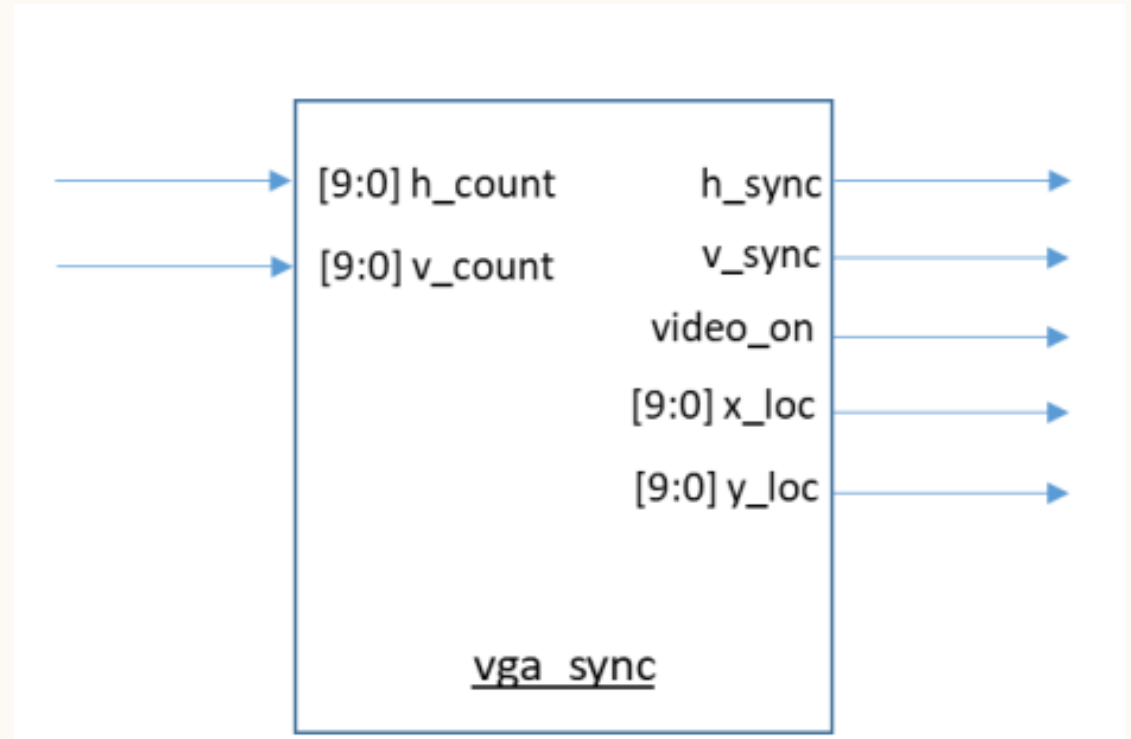
- During retrace, electron gun is off and no electrons are being shot (the beam is off), and the trace returns back quickly to monitor screen.
- ❖ During horizontal retrace, the electron beam moves from the right side of the display back to the left side, preparing to start drawing the next row.
- ❖ Vertical retrace occurs at the end of each frame, where the electron beam moves from the bottom of the display back to the top, preparing to start drawing the next frame.
- During the retrace periods, the VGA controller performs housekeeping tasks, such as adjusting internal registers, updating display settings, and synchronizing with the display's timing signals.

TO DISPLAY A PICTURE ON SCREEN

- Step 1: Generate the synchronization signals.
h_sync and *v_sync*
- Step 2: Provide pixel colour signals.
Red, Green, Blue

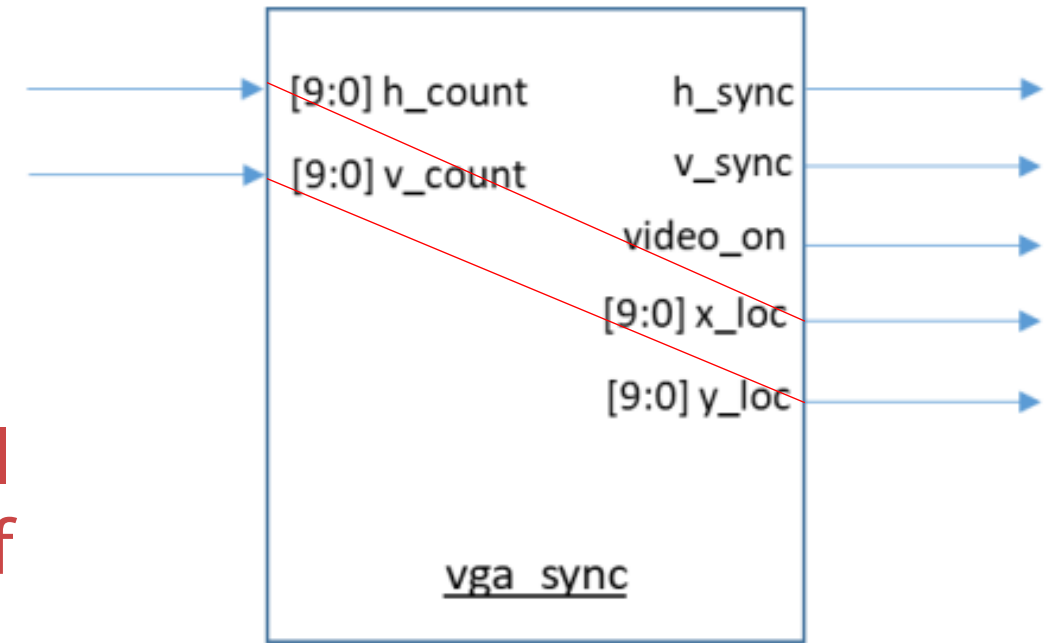
STEP 1: SYNCHRONIZATION CIRCUIT (VGA_SYNC)

- The *vga_sync* circuit generates timing and synchronization signals.
- It takes *h_count* and *v_count* as inputs and generate five outputs.



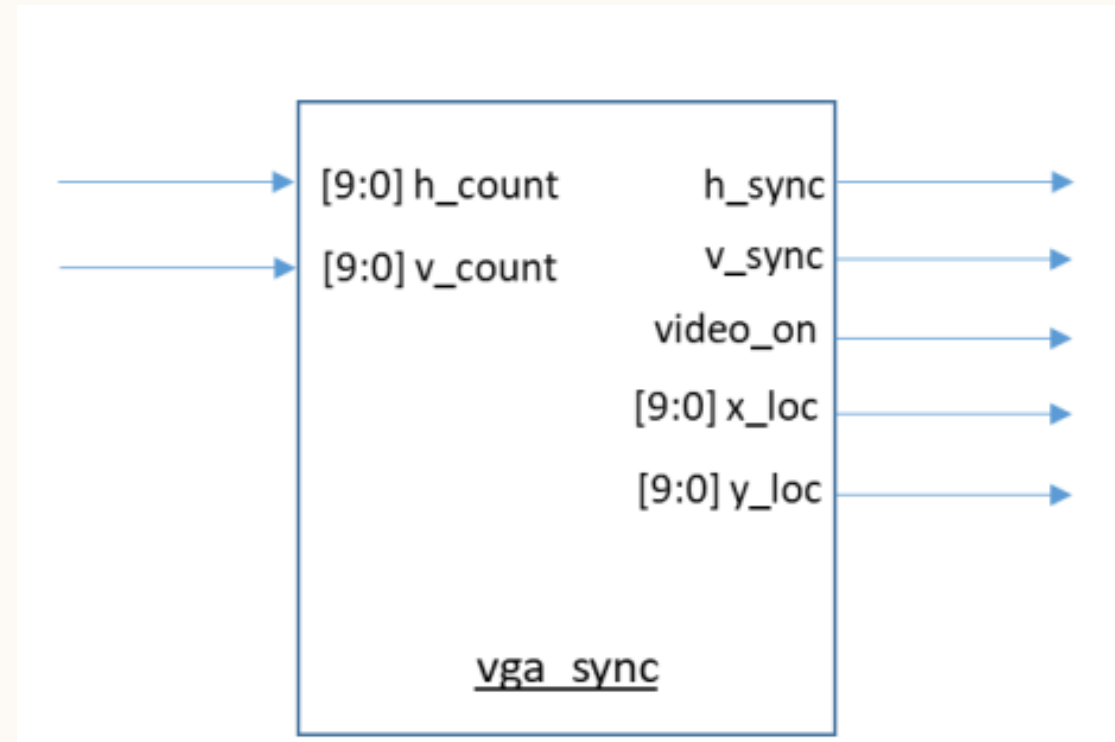
STEP 1: SYNCHRONIZATION CIRCUIT (VGA_SYNC)

- Specify the location of the current **pixels** (w.r.t to electron beam):
x_loc shows the location of pixel x
y_loc shows the location of pixel y
- The ***h_sync*** and ***v_sync*** signals are connected to the VGA port to **control the horizontal and vertical scans** of the monitor.



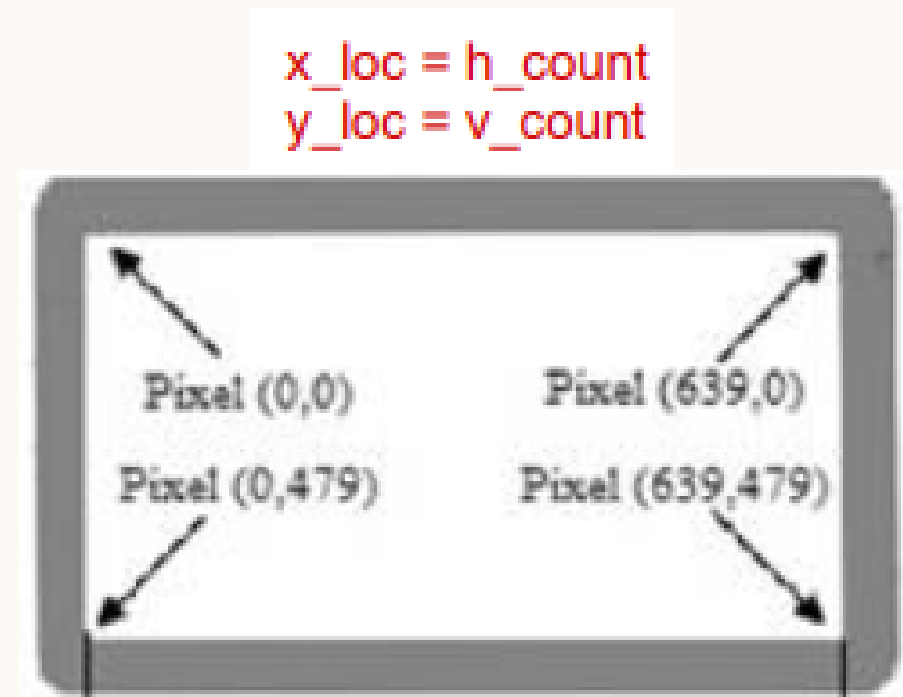
STEP 1: SYNCHRONIZATION CIRCUIT (VGA_SYNC)

- The *video_on* signal indicate **whether the current targeted pixel is in the displayable region.**
- *video_on* is asserted only when the *h_count* smaller than 640 and *v_count* is smaller than 480 - means when the beam is in the visible portion of the screen.

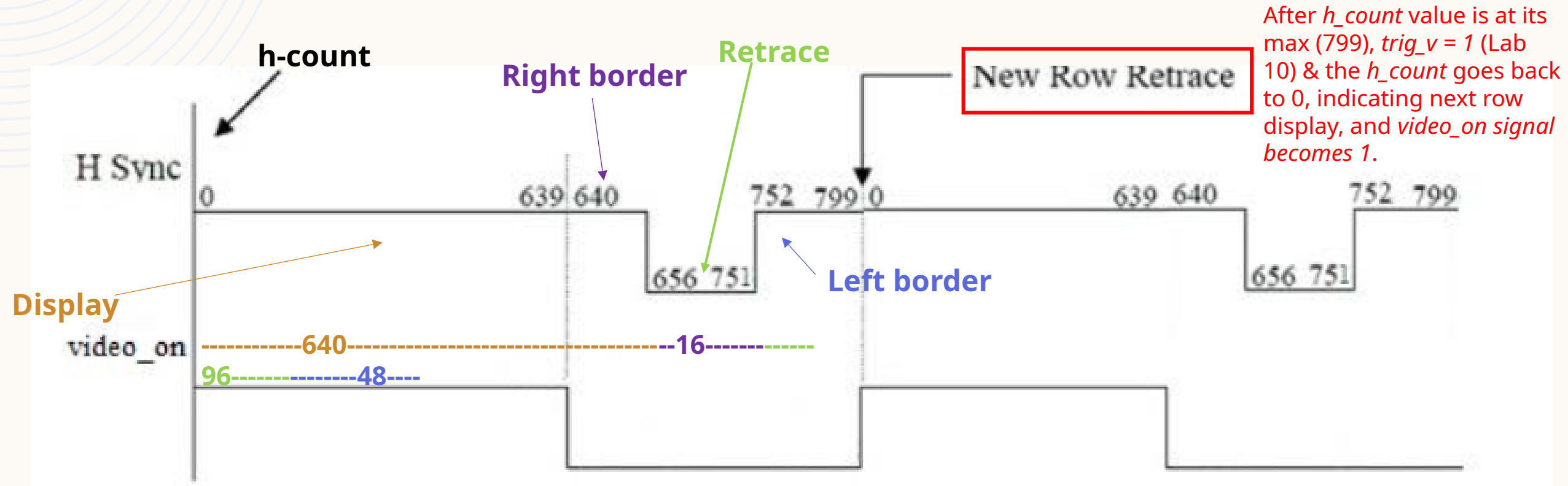


STEP 1: HORIZONTAL SYNCHRONIZATION (H-SYNC)

- During Horizontal Scan, electron beams move gradually from left to right, and then return to the right – *new row*.
 - Horizontal scan contains *800 pixels (0 - 799)*.
 - It can be divided into two regions:
 1. Display: 640 pixels
 $h_count = x_loc = 0 - 639$
 2. Blanking: It has 3 segments
 - a) Right Border (front porch): 16 pixels
 $h_count = x_loc = 640 - 655$
 - b) Retrace: 96 pixels
 $h_count = x_loc = 656 - 751$
 - c) Left Border (back porch): 48 pixels
 $h_count = x_loc = 752 - 799$
- Blanking Pixels: $16 + 96 + 48 = 160$ pixels (640 - 799)



HORIZONTAL SCAN: TIMING DIAGRAM

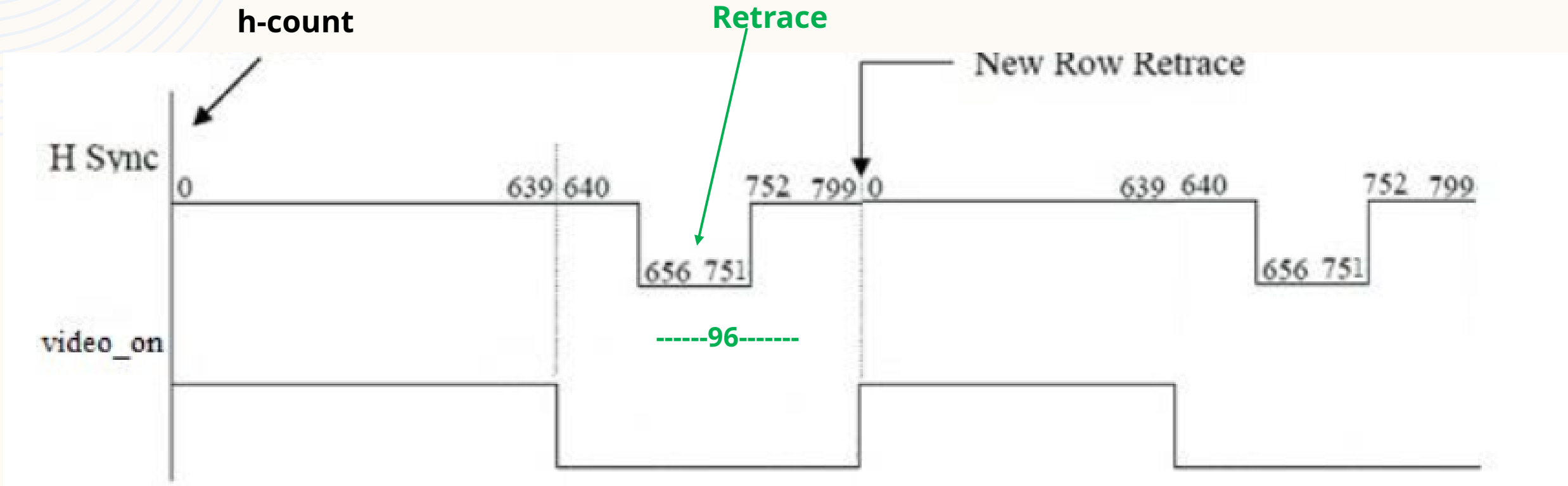


After h_count value is at its max (799), $trig_v = 1$ (Lab 10) & the h_count goes back to 0, indicating next row display, and $video_on$ signal becomes 1.

Horizontal Pixels:

- **H Display = 640**
DISPLAY: For h_count values 0 - 639 (Left to Right), $video_on = 1$ as current targeted pixel is in the displayable region.
- **H Front = 16**
- **H Retrace = 96**
- **H Back = 48**
BLANKING TIME: For h_count values 640 - 799, $video_on = 0$ as current targeted pixel is NOT in the displayable region.

HORIZONTAL SCAN: TIMING DIAGRAM



- $h_sync = 0$ only for h-counter values 656 - 751 (consists of 'Retrace' = 96 pixels) indicates the beginning of a new scan row.
- During this time, the electron beam in the display device returns from the right side of the screen to the left side to begin drawing the next scan line.

STEP 1: VERTICAL SYNCHRONIZATION (V-SYNC)

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- During Vertical scan, electron beams move gradually from top to bottom, and then return to the top – *new screen/frame*.
- The time unit of the movement is represented in terms of horizontal scan *lines*.
- Vertical scan contains 525 *lines* (0 - 524).

- It can be divided into two regions:

1. Display: 480 lines

$v_count = y_loc = 0 - 479$

2. Blanking: It has 3 segments

a) Bottom Border (front porch): 10 lines

$v_count = y_loc = 480 - 489$

b) Retrace: 2 lines

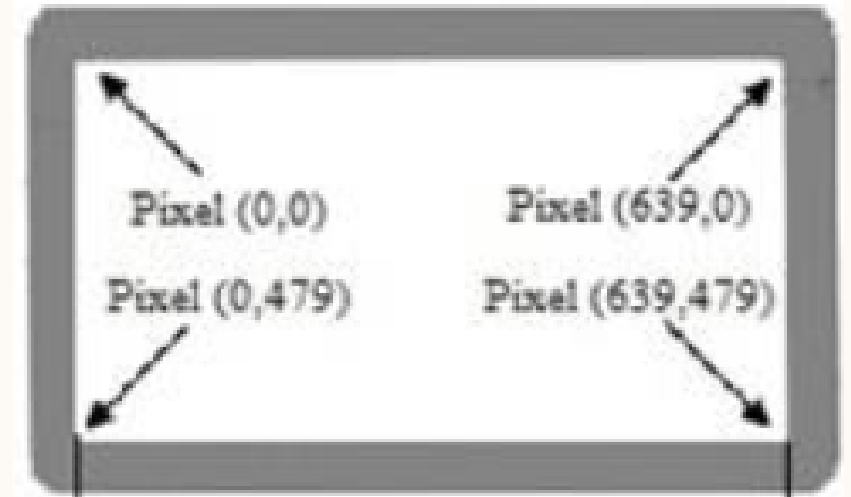
$v_count = y_loc = 490 - 491$

c) Top Border (back porch): 33 lines

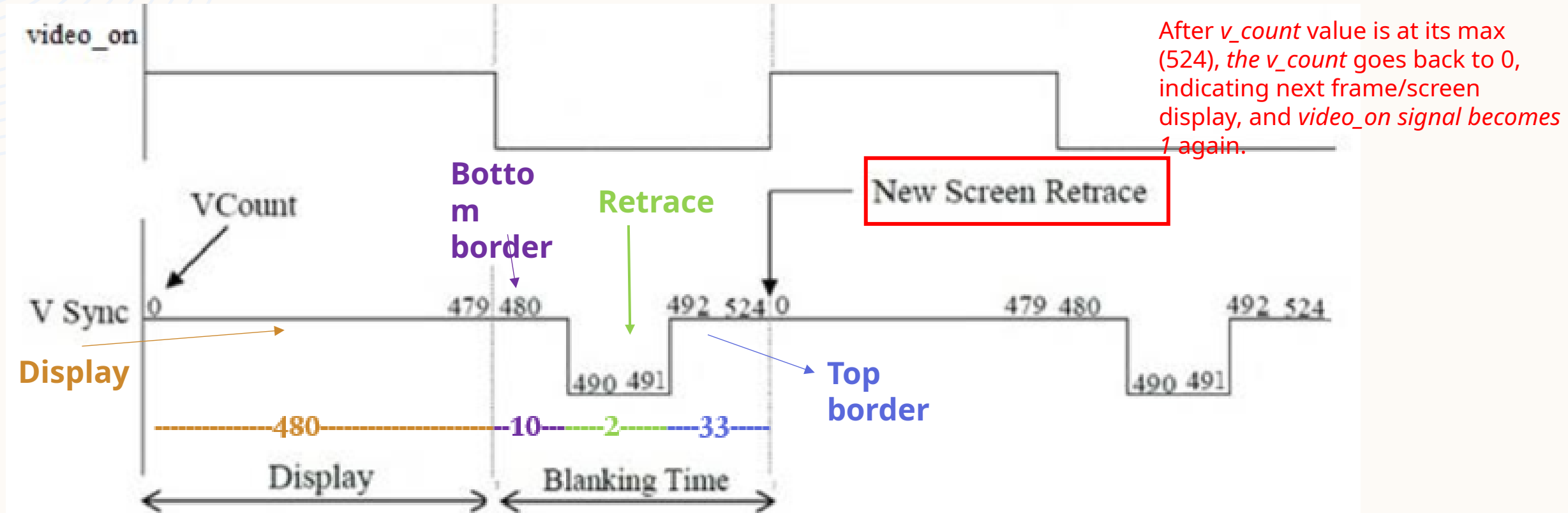
$v_count = y_loc = 492 - 524$

Blanking Lines: $10 + 2 + 33 = 45$ lines (480 - 524)

$x_loc = h_count$
 $y_loc = v_count$



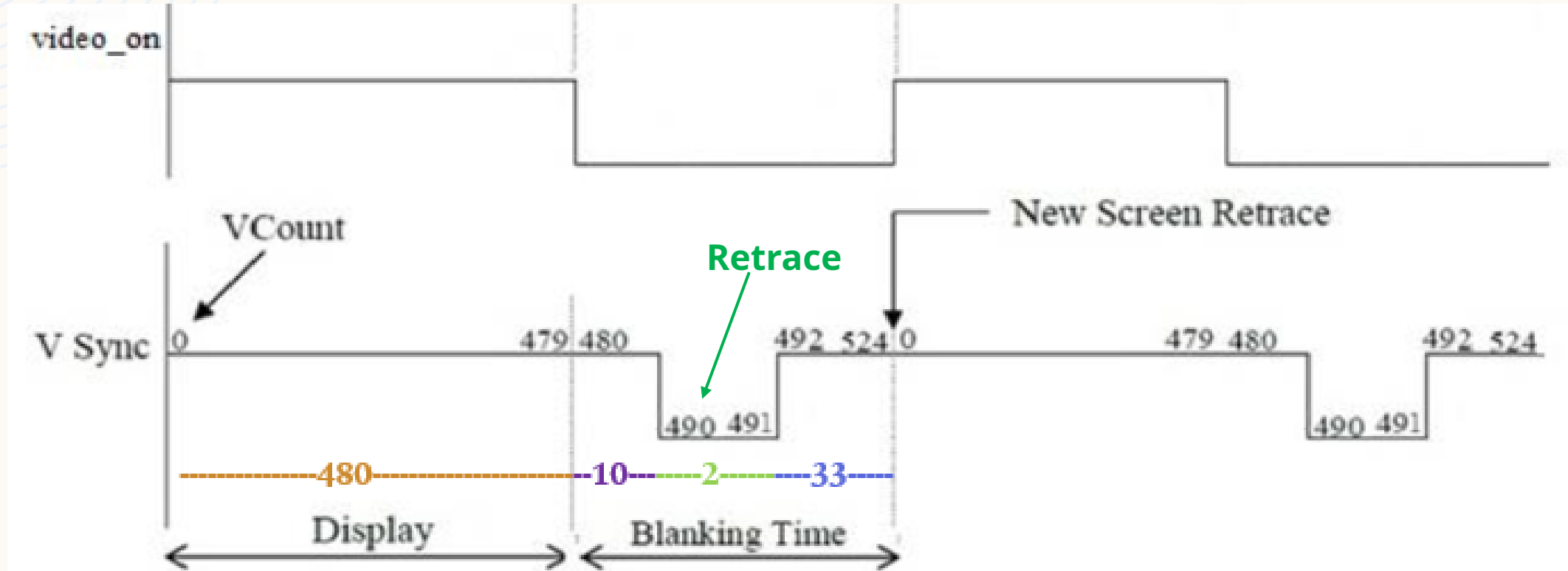
VERTICAL SCAN: TIMING DIAGRAM



Vertical Lines:

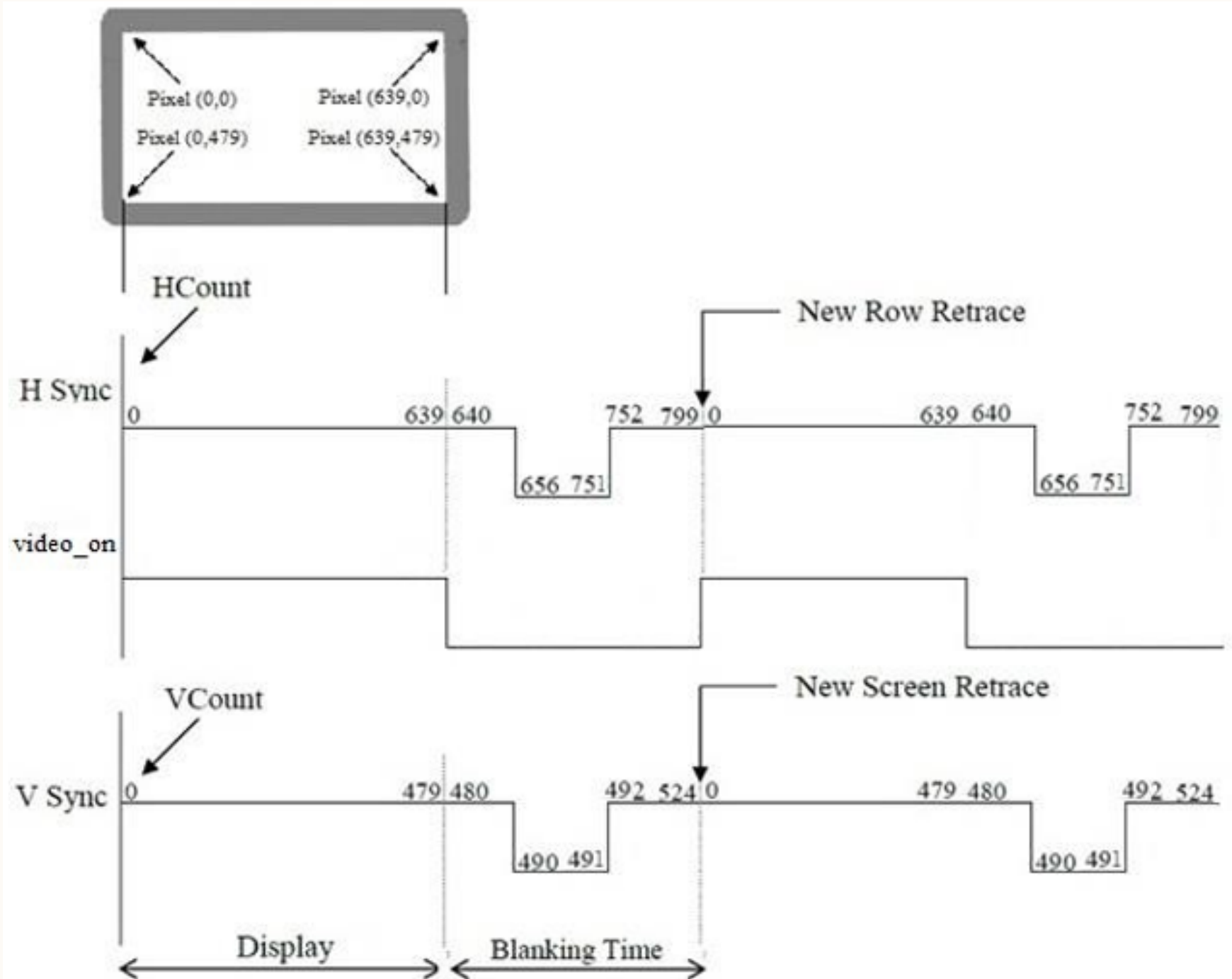
- **V Display = 480**
DISPLAY: For v_count values 0 - 479 (Top to Bottom), $video_on = 1$ as current targeted pixel is in the displayable region.
- **V Front = 10**
- **V Retrace = 2**
- **V Back = 33**
BLANKING TIME: For (v_count values 480 - 524, $video_on = 0$ as current targeted pixel is **NOT** in the displayable region.

VERTICAL SCAN: TIMING DIAGRAM

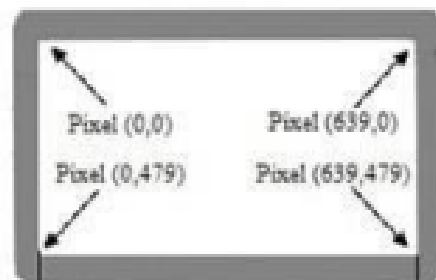


- $v_sync = 0$ for v-counter values 490 - 491 (consists of 'Retrace' = 2 lines) indicates the beginning of a new scan frame.
- During this time, the electron beam in the display device returns from the bottom of the screen to the top to begin drawing the next frame.

V-SYNC MODULE – TIMING DIAGRAM



$x_loc = h_count$
 $y_loc = v_count$



BLANKING TIME:
 For h_count values 640 - 799 (horizontal pixels resolution), $video_on = 0$ as current targeted pixel is NOT in the displayable region.

DISPLAY:
 After h_count value is at its max (799), $trig_v = 1$ (Lab 10) & the h_count goes back to 0, indicating next row display, and $video_on$ signal becomes 1.

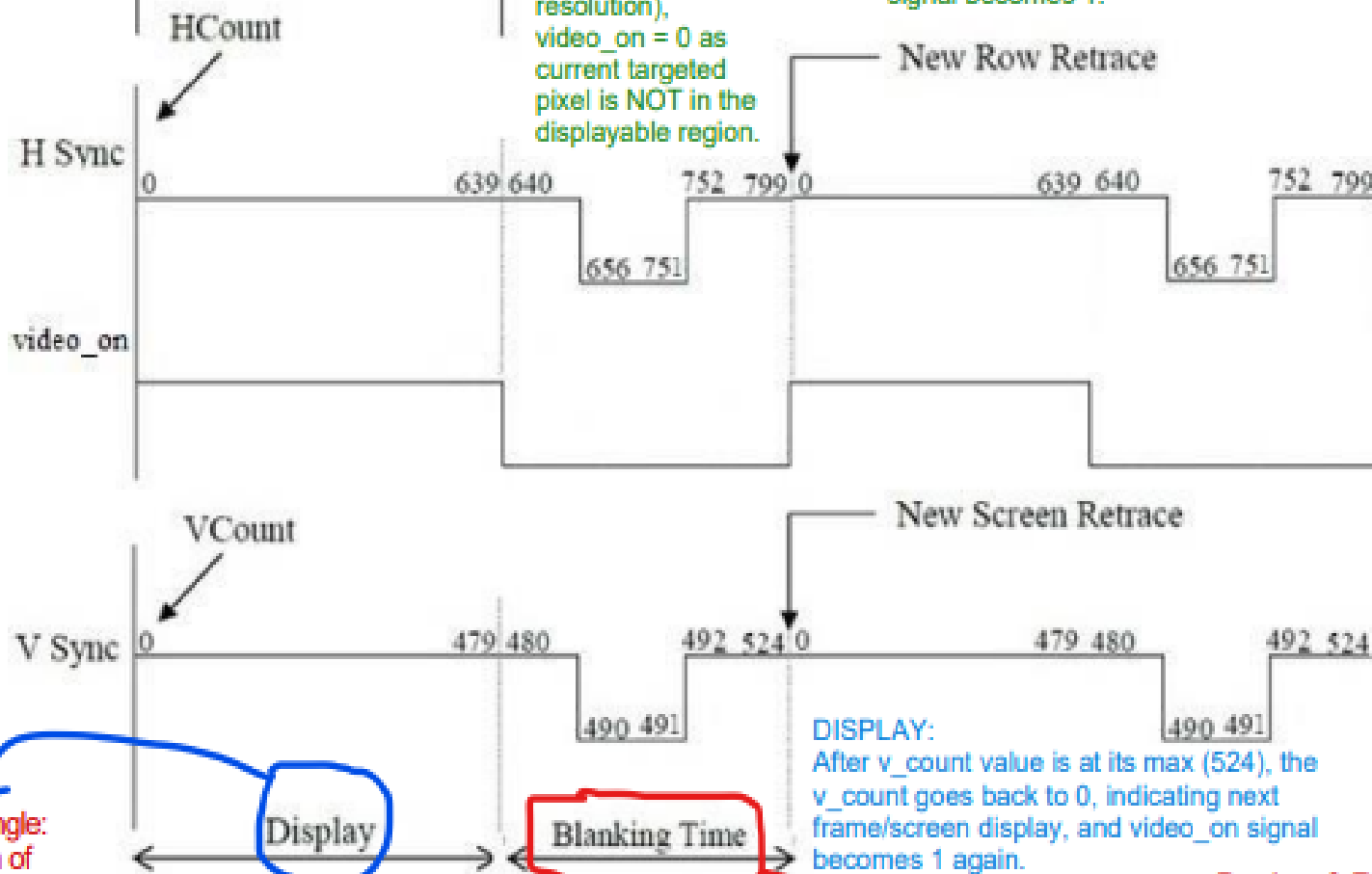


Figure 11. 3

PURPOSE OF BLANKING TIME

1. Synchronization:

The blanking time provides a period during which the VGA controller can synchronize (horizontal and vertical signals) its timing with the display hardware to ensure proper alignment of the image on the screen.

2. Horizontal and Vertical Retrace:

The electron beam returns to the starting position of the next scan line or frame to maintain the correct timing and alignment of the displayed image.

3. Prevention of Visual Artifacts:

Tearing or flickering on the display due to rapid changes in the displayed image.

4. Overhead for Hardware Operations:

For efficient operation and management of the display system such as adjusting internal registers, updating display settings, and performing housekeeping tasks within the VGA controller.

TASK A – VGA SYNC MODULE

```

“timescale 1ns / 1ps
module vga_syne (
input [9:0] h_count,
input [9:0] v_count,
output h_sync,
output v_sync,
output video_on,      // active area
output [9:0] x_loc,    // current pixel x- location
output [9:0] y_loc     // current pixel y-location
);

// horizontal
localparam HD =      // Horizontal Display Area
localparam HF =      // Horizontal (Front Porch) Right Border
localparam HB =      // Horizontal (Back Porch) Left Border
localparam HR =      // Horizontal Retrace

// vertical
localparam VD =      // Vertical Display Area
localparam VF =      // Vertical (Front Porch) Bottom Border
localparam VB =      // Vertical (Back Porch)Top Border
localparam VR =      // Vertical Retrace
endmodule           // vga_syne

```

- Write down the values of the parameters to define limits.
- Write down the logic for *h_sync*, *v_sync*, *video_on*, *x_loc* and *y_loc* outputs.

- Hint:

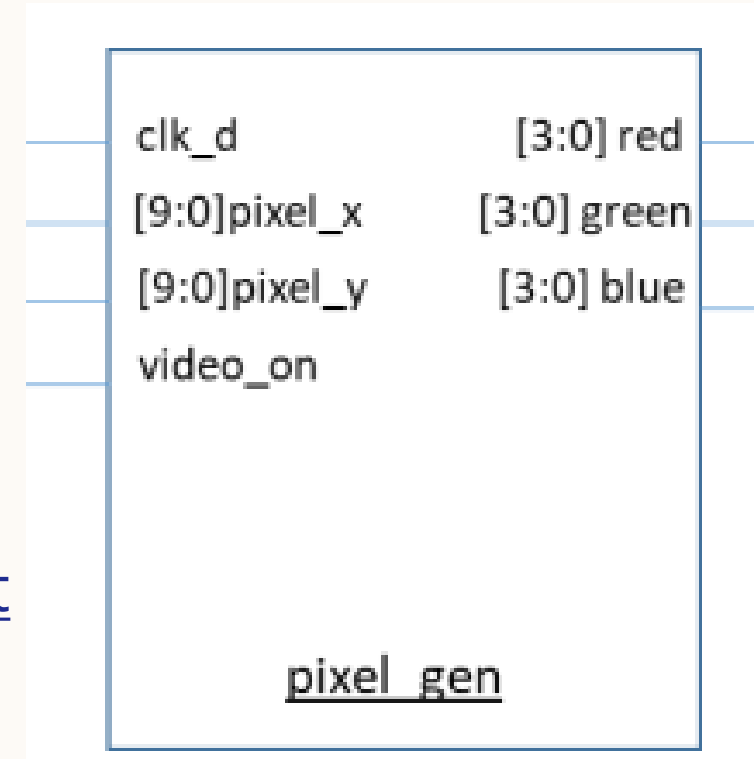
- For what values of which param *h_sync* == 1, and for what values of which param *h_sync* == 0 (refer to the Figure 11.3 in the manual).
- assign *h_sync* = (*h_count* < HD+HF) || (*h_count* >= HD + HR + HF) ? 1:0;
//h_sync == 1 for h_count == 0 to 655 (HD + HF = 655) or h_count = 752 to 799 (HD + HR + HF = 751)

localparam **COMMAND**

- Localparam prevents the values to be overwritten (directly) from outside the module.
- Once the variables are declared with 'localparam' the values stays constant.
- When the module is instantiated within another module the values can't be passed for these data types.
- The localparam in Verilog is a kind of constant that is local for modules and available only inside the module in which it is declared.

STEP 2: PIXEL GENERATION CIRCUIT

- The pixel generation circuit generates the three 4-bit color signals (RGB signals).
- A color value is obtained according to the current coordinates of the pixel (the *x_loc* and *y_loc* signals), and the external control (*clk_d*) and data signals (*video_on*).
- This module has one input *clk_d* (of 25MHz), two 10-bit input *pixel_x* and *pixel_y* and one input *video_on*.
- This module generates three 4-bit signals of *Red*, *Green* and *Blue color*.



THREE-BIT VGA COLOR COMBINATIONS

Red (R)	Green (G)	Blue (B)	Resulting color
0000	0000	0000	black
0000	0000	1111	blue
0000	1111	0000	green
0000	1111	1111	cyan
1111	0000	0000	red
1111	0000	1111	magenta
1111	1111	0000	yellow

TASK B - PIXEL GENERATION CODE

```

`timescale 1ns /1ps
module pixel_gen(
input clk_d, // pixel clock
input [9:0] pixel_x,
input [9:0] pixel_y,
input video_on,
output reg [3:0] red=0,
output reg [3:0] green=0,
output reg [3:0] blue=0
);
always @(posedge clk_d)
begin
    if ((pixel_x == 0) || (pixel_x == 639) || (pixel_y == 0) ||
        (pixel_y == 479))
        begin
            red <= 4'hF; green <= 4'hF; blue <= 4'hF;
        end
    else
        begin
            red <= video_on ? (pixel_y > 240 ? 4'hF : 4'h0) : (4'h0);
            green <= video_on ? (pixel_y > 240 ? 4'h0 : 4'hF) :
(4'h0);
            blue <= 4'h0;
        end
    end
end

```

For Border:

- Sets the color of pixels located at the edges of the display area to **white**, indicating the boundary of the display.
- This is achieved by detecting when the pixel coordinates (pixel_x and pixel_y) are at the edges and assigning maximum intensity (4'hF) to all color components (red, green, and blue) for those pixels.

For the other part of the display screen:

• For the red signal:

If video_on is true:

If pixel_y is greater than 240, set red to its maximum intensity (4'hF), indicating a fully red pixel.

Otherwise, set red to zero intensity (4'h0), indicating no red color.

• For the green signal:

If video_on is true :

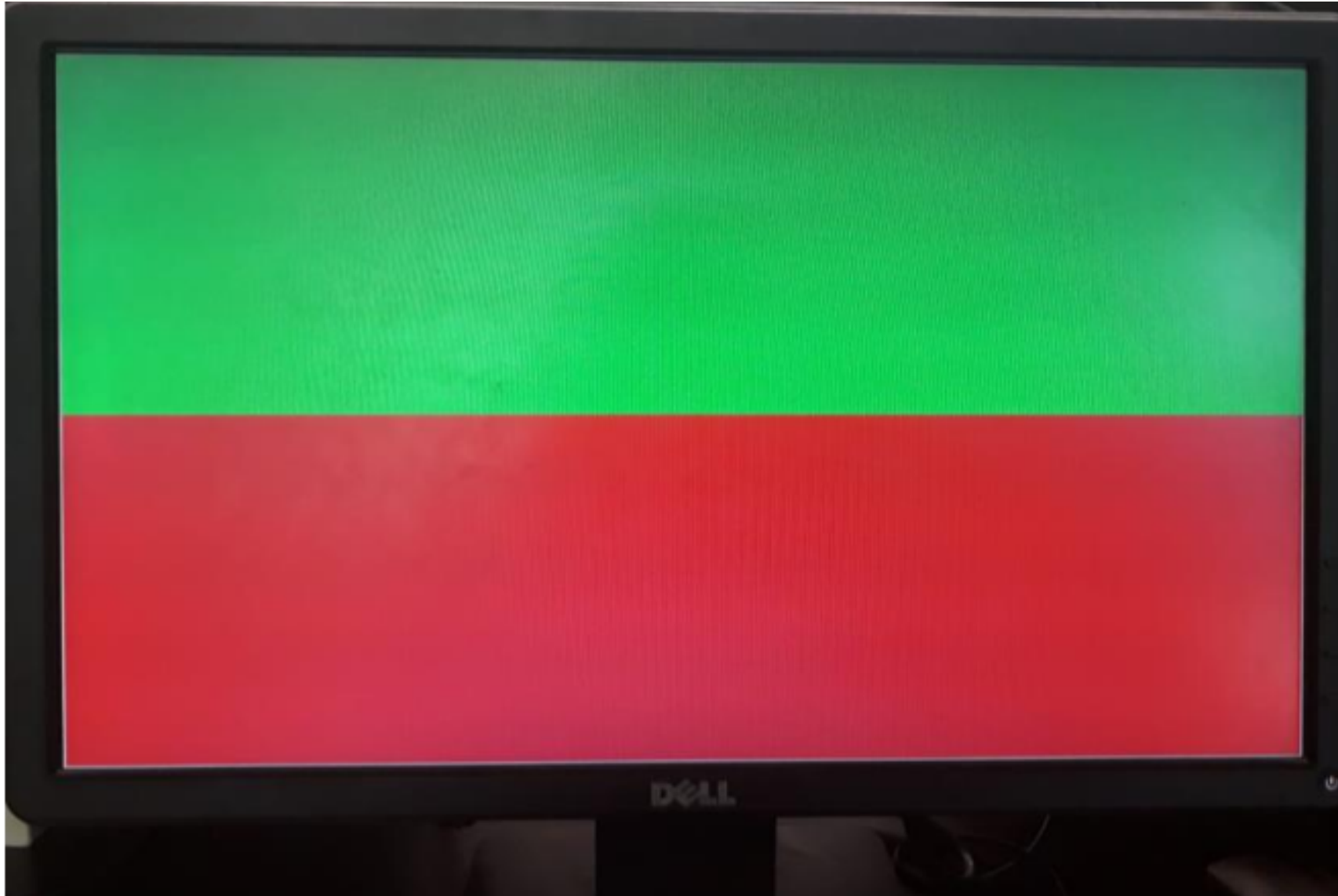
If pixel_y is greater than 240, set green to zero intensity (4'h0), indicating no green color.

Otherwise, set green to its maximum intensity (4'hF), indicating a fully green pixel.

• For the blue signal:

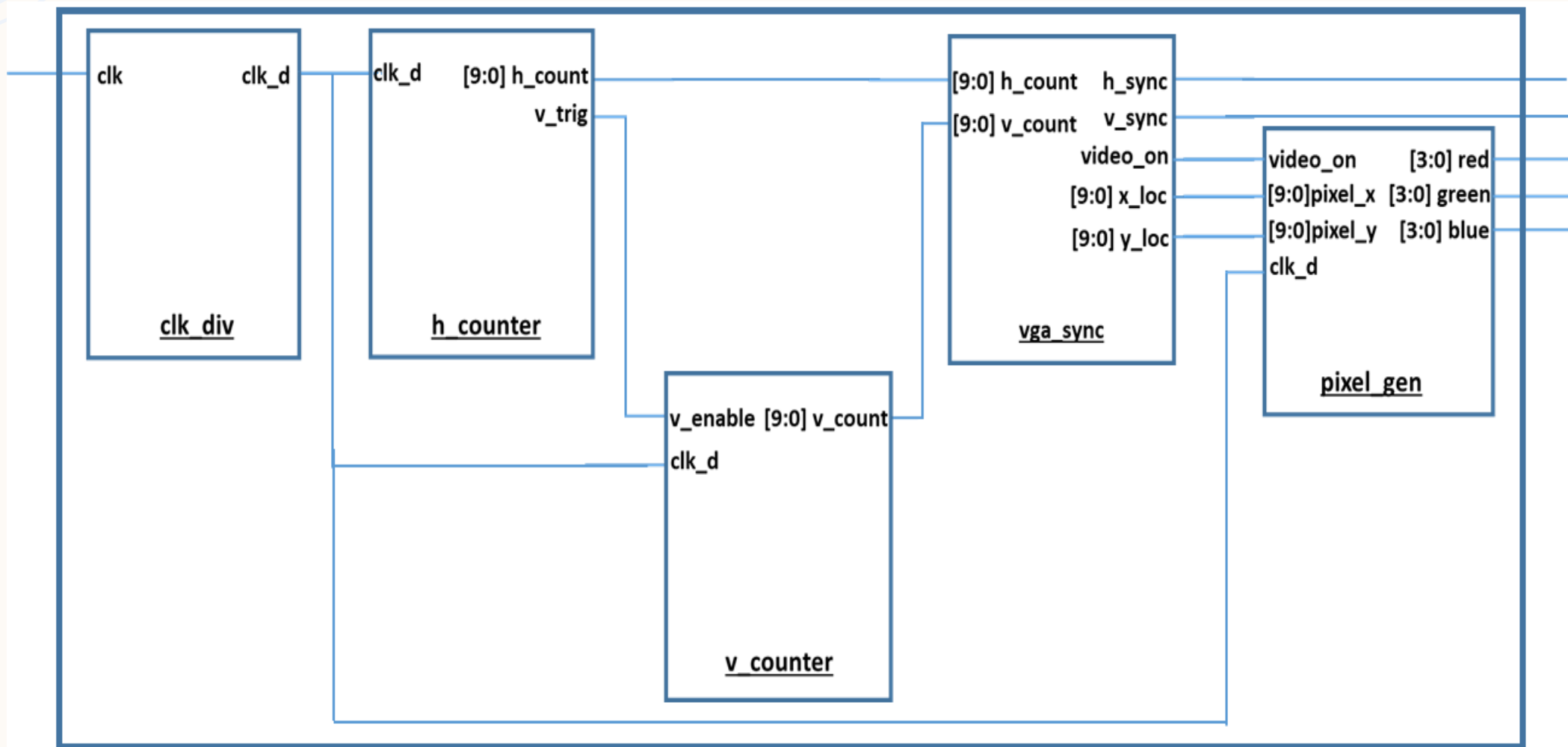
Set blue to zero intensity (4'h0), indicating no blue color.

TASK B - PIXEL GENERATION OUTPUT

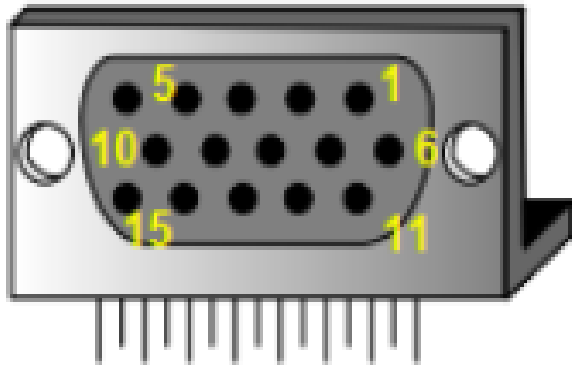


TASK C - TOP LEVEL MODULE

Top_Level module receives a 100Mhz Clock as input and generate 1-bit *h_sync* and *v_sync* signals, and three 4-bit signals for *Red, Green and Blue* color.

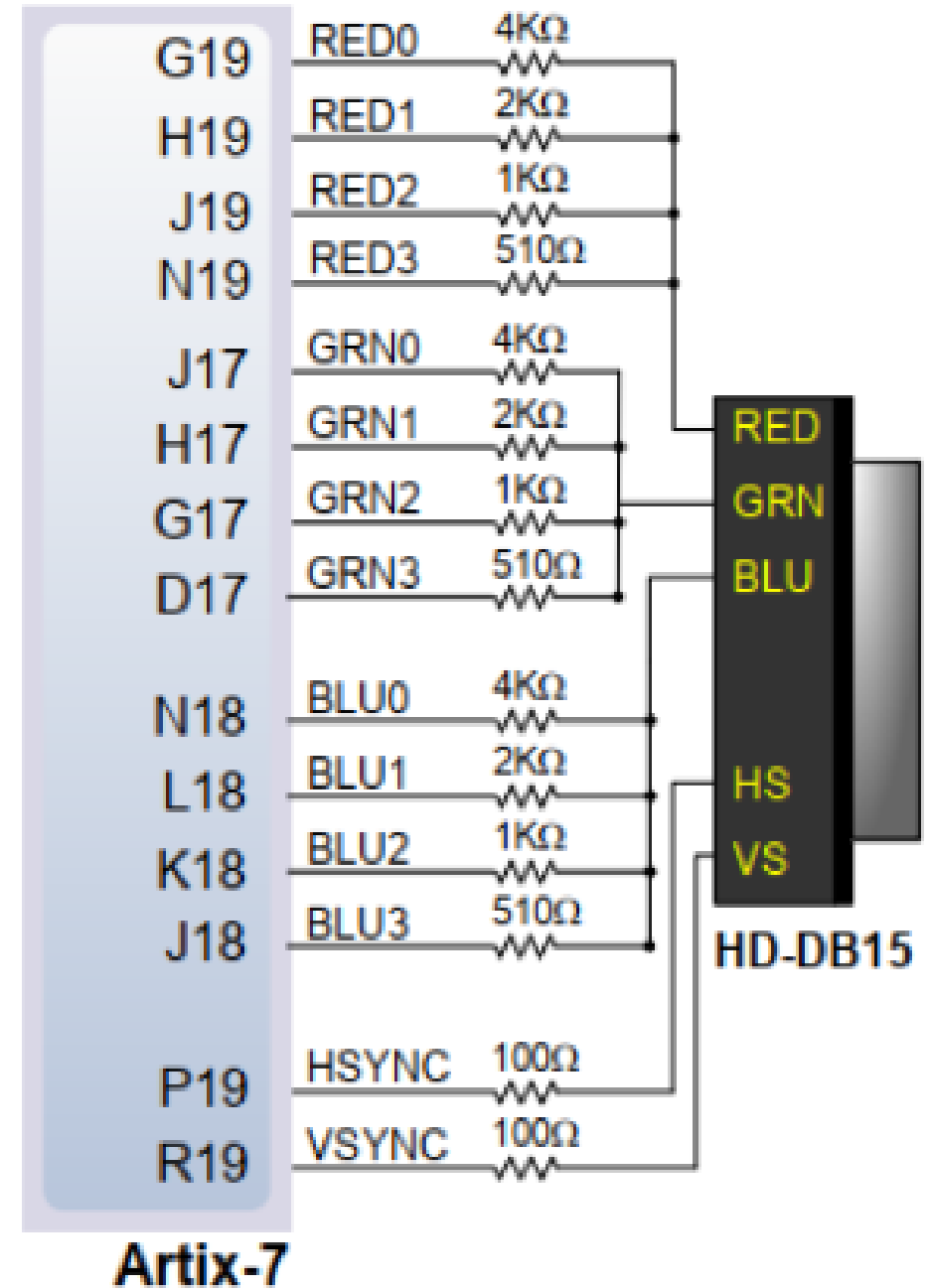


PIN CONFIGURATIONS



Pin 1: Red	Pin 5: GND
Pin 2: Grn	Pin 6: Red GND
Pin 3: Blue	Pin 7: Grn GND
Pin 13: HS	Pin 8: Blu GND
Pin 14: VS	Pin 10: Sync GND

The Basys 3 board includes a single 100 MHz oscillator connected to pin W5 - clock.



V_COUNTER CODE (IF ERROR IN SYNTHESIS)

```
always @(posedge clk)
```

```
begin
```

```
if ((enable_v ) &(v_count<524))
```

```
v_count <= v_count+1;
```

```
end
```

```
else if (enable_v )
```

```
v_count<=0; // non-blocking
```

```
end
```

```
end
```

- For VGA tasks, disconnect the VGA Cable from the PC's CPU, and connect it to the Basys-3 FPGA Board.
- You must show the VGA result for *Task c & Exercise* to RA.